BE IT 2ND YEAR, 2ND SEMESTER, 2021

OBJECT ORIENTED SYSTEMS

Time: 2 hours **Full Marks: 30**

Question 1: [CO1]

a) Show with a suitable program how the main() method can be overloaded. Provide 3 overloaded versions to demonstrate. Show how each of the functions can be invoked.
Can we overload two methods having the signatures as ***public int sum(int a, int b)*** and ***public static int sum (int x, int y)***? If your answer is YES, show how it can be done using suitable code snippet; if your answer is NO, provide supporting reasons for it.

b) What is the purpose of using interfaces in Java?
Consider 2 interfaces IN1 and IN2 each having a *display()* method of its own. Now create a class *OOS* implementing these two interfaces. Show how the *display()* method of the interface IN1 can be invoked using an object of the *OOS* class.

c) What is the purpose of declaring a method as *final* in a class? Show whether (and how) a private member method of a class can be overridden.

[(4+3)+(1+3)+(1+3)=15]

Question 2:

[CO1]

a) What do you mean by static import of packages? Show its utility with suitable examples.
Show how you can create a sub-package.

b) Write a suitable program for the following scenario.
Consider a case of multi-level inheritance where a class *Base* having a parameterized constructor and a member method *fun()*is inherited by the class *Child*. The *Child* class has its own parameterized constructor and its own definition of the overridden *fun()* method. Show how an object of class *GrandChild* inherited from *Child* can invoke the *Base* class constructor and the *fun()* method.

[CO3]

c) Define a generic class with a member method *find(A,P)* that take sorted arrays ***A*** of different datatypes (such as int, float, char) as input and checks whether the element ***P*** is present (and at what position) within the array or not using binary search technique.

[(2+2)+5+6=15]